Labour Demand 2: Non-wage Benefits and Quasi-Fixed Costs

EC306- Wages and Employment

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Winter 2026

Goals of This Section

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Today we will:

- Distinguish between workers and hours as an input into production
- Incorporate non-wage benefits into the labour demand model
- Add costs that are independent of hours worked (quasi-fixed costs)
- Discuss the implications of quasi-fixed costs for labour demand

Background

Background

Previous Assumption

Labour is a purely variable input paid a wage per hour worked

In reality it is not so simple:

- Workers are sometimes paid other benefits besides wages
  - CPP, EI, retirement benefits, paid vacation, sick leave, etc.
- There are costs beyond the hourly wage
  - Hiring and training costs
  - Costs of dismissal
  - Non-wage benefits described above

These benefits and costs complicate the labour demand decision

Non-Wage Benefits

Non-wage Benefits

Compensation typically includes more than just wages

Common non-wage benefits:

- Health insurance (beyond public health care)
- Retirement benefits
- Paid vacation and sick leave
- Other perks (gym memberships, childcare, etc.)
- These benefits have value to workers
- Increased as a share of total compensation through the 90s and early 2000s

Non-wage Benefits

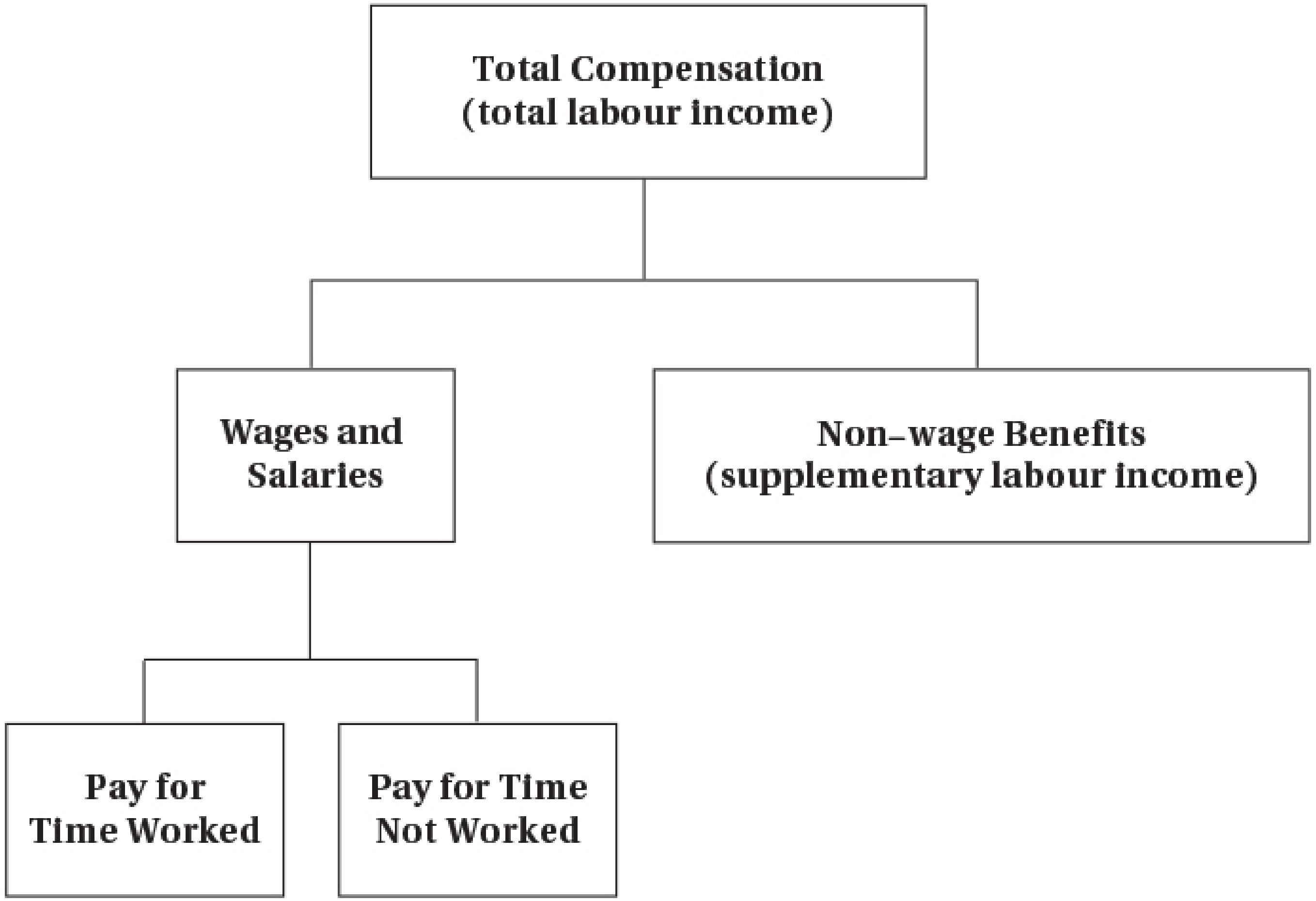


FIGURE 6.1 Components of Total Compensation

Total compensation includes wages plus non-wage benefits. The wage and salary component can be further divided into the part directly corresponding to time worked and that which does not depend on time worked (such as vacation pay).

Non-wage Benefits

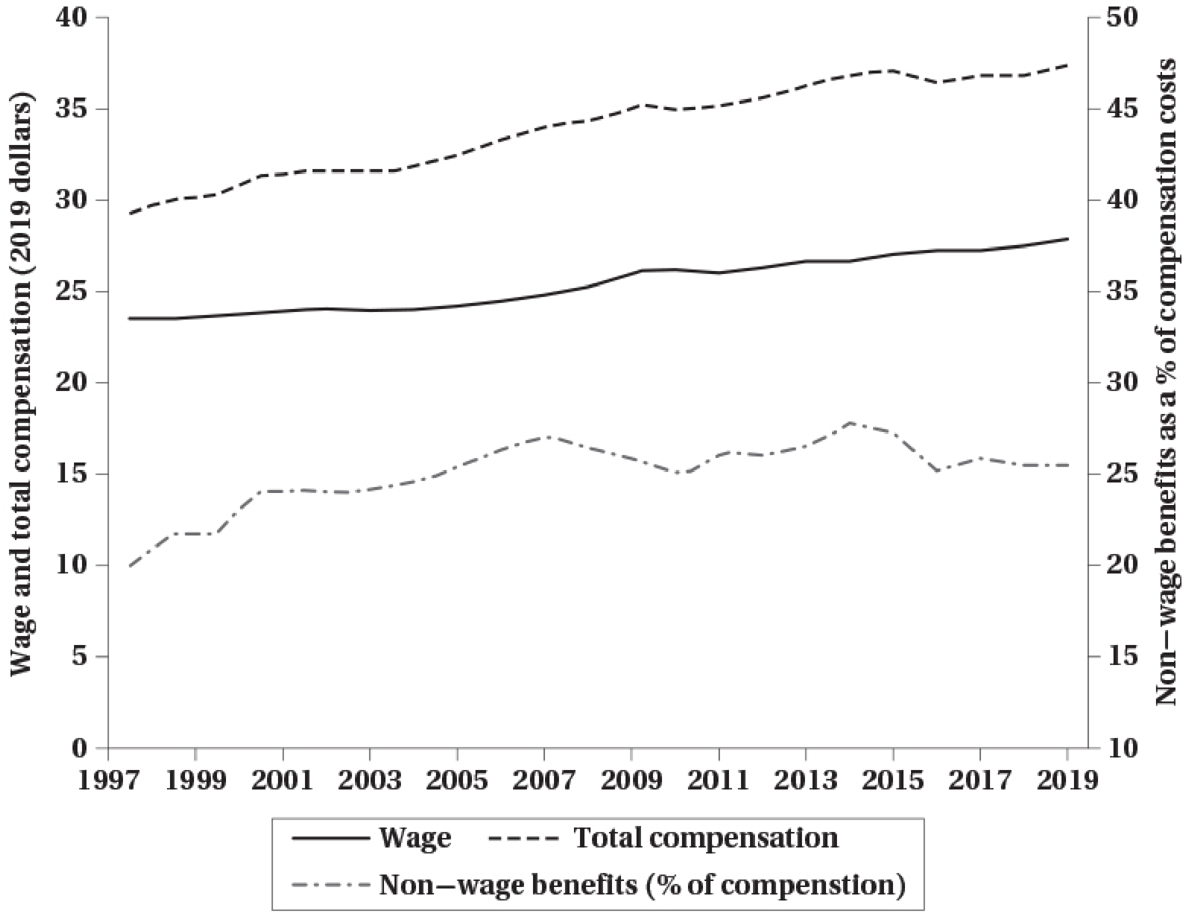


FIGURE 6.2 Total Compensation, Wage, and Non-wage Benefits

This figure shows the evolution of total compensation per hour, wages per hour, and non-wage benefits as fraction of total compensation. Total compensation and wages are expressed in dollars of 2019. Total compensation if growing faster than wages, reflecting the growing contribution of non-wage benefits to total compensation.

SOURCE: Statistics Canada table 36-10-0489-01 (total compensation from the System of National Accounts) and table 14-10-0064-01 (hourly wage from the Labour Force Survey).

Non-wage Benefits

Why do firms pay benefits instead of wages?

- Tax advantages: sometimes benefits are not taxed
- Economies of scale: pensions and insurance are cheaper when provided to a group (avoids adverse selection)
- Worker preferences: risk-averse workers may prefer health insurance to wages
- Legal requirements

Non-wage Benefits

More reasons:

- Better firm planning: benefits like sick leave reduce absenteeism and provide contingency plans
- Productivity: subsidized meals keep people working
- Deferred compensation: retirement benefits help retain workers
  - Workers may accept lower wages in exchange for future benefits
  - Acts as a screen for good workers

Non-wage Benefits

Governments also prefer and mandate them:

- Firm pensions reduce need for public pensions
- Better health outcomes reduce public health care costs
- Improve social safety net through sick leave and unemployment insurance

Tip

With all these reasons, it is not surprising that non-wage benefits are common

Quasi-fixed Costs

Introduction

> **Important Distinction**
>
> Some costs are incurred per worker regardless of hours worked. Others are incurred per hour worked.

Quasi-fixed costs: costs that occur per worker but are relatively independent of hours worked

- Pure quasi-fixed costs do not vary with hours worked at all
- In practice, many costs are partially quasi-fixed
- Some recurring (e.g. monthly EI premiums), others one-time (e.g. hiring and training costs)

Examples of Quasi-fixed Costs

Hiring costs:

- Recruiting, interviewing, and training are costly
- Incurred when a worker is hired regardless of hours worked
- Firms generally try to minimize hiring costs by retaining workers

Termination costs:

- Severance pay, legal fees, and administrative costs
- Initially depend on time with company, but have a maximum so become fixed
- Firms may be reluctant to lay off workers because of these costs

Examples of Quasi-fixed Costs

Employer contributions to payroll taxes:

- EI and CPP are initially related to work hours
- But they have a ceiling so become quasi-fixed for high hours worked

Other non-wage benefits:

- Employer contributions to life insurance, medical, dental plans
- Often have a fixed component regardless of hours worked

> **Note**
>
> If a cost always varies with hours worked it is **not** quasi-fixed — e.g. hourly wage, overtime premiums, per-hour health and safety costs

Effects of Quasi-fixed Costs

Previously: labour demand determined by equating MRP_L to the wage rate

With quasi-fixed costs, the cost of hiring a worker is higher. There is also a dynamic component — firms amortize costs over time:

- Firms may pay high training costs now
- But expect the worker to remain to recoup those costs

A profit maximizing firm will hire workers up to the point where:

$$H + T + \sum_{t=1}^{N} \frac{w_t}{(1+r)^t} = \sum_{t=1}^{N} \frac{VMP_t}{(1+r)^t}$$

Where H = hiring cost, T = training cost, w_t = wage, VMP_t = value of marginal product, r = discount rate, N = expected tenure

Effects of Quasi-fixed Costs

This equation has several implications:

- Firms reluctant to hire workers for short periods if hiring costs are high
- Wage does not have to equal marginal product in each period
- Future productivity matters, but is discounted
- Firms may pay more than current marginal product if future productivity is expected to be high

When hiring costs are a factor, future productivity must exceed the total wage:

$$\sum_{t=1}^{N} \frac{VMP_t}{(1+r)^t} > \sum_{t=1}^{N} \frac{w_t}{(1+r)^t}$$

Effects of Quasi-fixed Costs

Like Any Other Investment Decision

Firms invest in workers when the present value of future returns exceeds the present value of costs

- Hiring and training costs → “upfront” investment costs
- Wages → “ongoing” investment costs
- Future productivity → returns on investment
- You only make the investment if you expect to earn at least as much as you spend

Effects of Quasi-fixed Costs

	Costs					
Employment Duration (N)	Fixed (H + T)	Variable ($\sum W_t$)	Total	Benefits ($\sum VMP_t$)	Total Profit	Yearly Profit
1 year	\$15,000	\$25,000	\$40,000	\$30,000	−\$10,000	−\$10,000
2 years	15,000	50,000	65,000	60,000	−5,000	−2,500
3 years	15,000	75,000	90,000	90,000	0	0
4 years	15,000	100,000	115,000	120,000	5,000	1,250
5 years	15,000	125,000	140,000	150,000	10,000	2,000

Things Explained by Quasi-Fixed Costs

Overtime

Firms sometimes work employees longer despite higher overtime wages

Explanation

High quasi-fixed costs of hiring additional workers make it cheaper to pay overtime than incur hiring and training costs — especially true for high-skill workers requiring a lot of training

Temp and Gig Workers

Firms do not always hire part- or full-time employees directly

They sometimes use temp agencies or “gig” workers:

- Laurier hires a cleaning firm and does not employ many janitors directly
- Uber classifies drivers as independent contractors

Explanation

By doing this, firms avoid paying quasi-fixed costs

Layoffs

During downturns firms sometimes keep workers around

Explanation: Labour Hoarding

- Paying severance and legal fees may be costly
- Firms may prefer to keep workers on payroll if they expect a quick recovery
- Firms reluctant to lay off skilled workers due to high training costs

This also explains why highly educated workers tend to suffer less unemployment during recessions

Layoffs

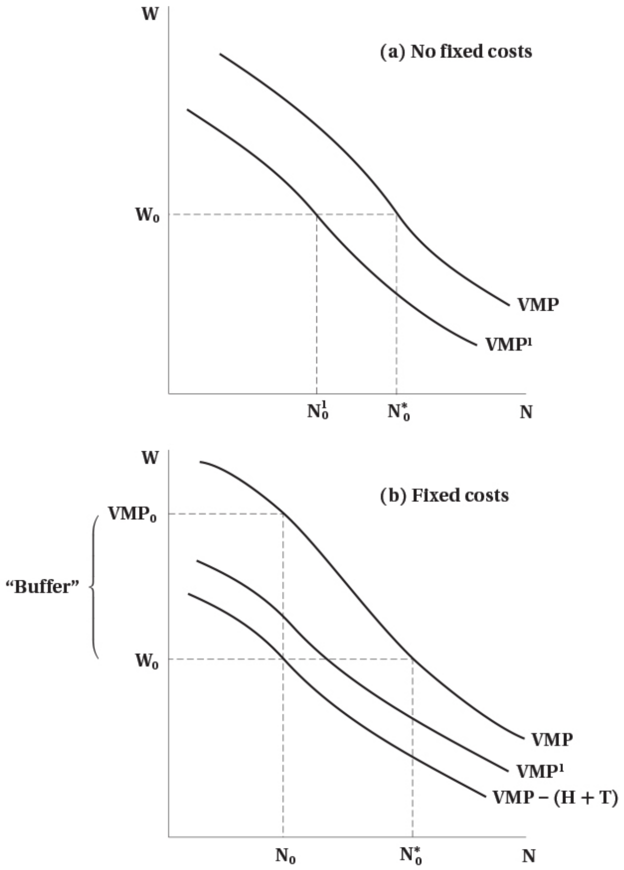


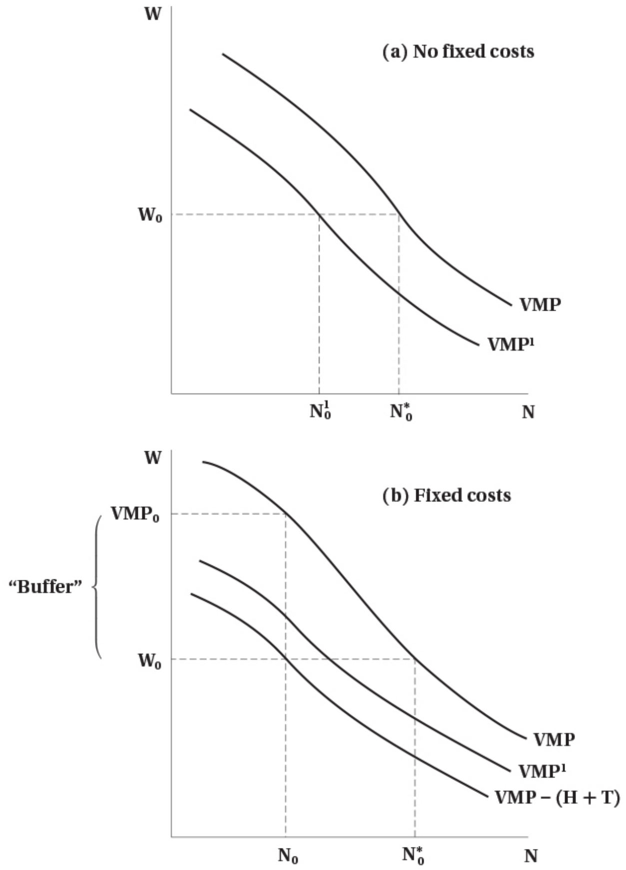
FIGURE 6.3 Nonrecurring Fixed Employment Costs and Changes in Labour Demand

In panel (a), a decline in market conditions leads to an inward shift of the VMP schedule, from VMP to VMP^1. The profit-maximizing level of employment falls from N_0^* to N_0^1 at the prevailing wage of W_0. By contrast, if there are fixed costs of hiring workers, employment may not fall as much. First, the original level of employment will be at N_0, not N_0^*, as employers choose employment so that the wage equals the VMP net of hiring and training costs, VMP − (H + T). At N_0, the VMP (VMP_0) exceeds W_0 by a "buffer," H + T. As long as the new $VMP^1 > VMP - (H + T) = W_0$, it will not be optimal for the firm to change its employment level from N_0.

Top panel — without quasi-fixed costs:

- Firm hires to point where $VMP = w$
- In downturn, VMP shifts down (lower demand for output)
- With same wage, firms hire fewer workers

Layoffs



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Bottom panel — with quasi-fixed costs:

- Quasi-fixed costs shift VMP curve down prior to downturn
  - Firm hires to point where $VMP - (H + T) = w$
- Creates wedge between VMP and wage
- After downturn, VMP shifts down
  - But because of wedge, firm may not change employment

Segmentation

Hiring costs mean trained and untrained employees are different

- Firms reluctant to hire untrained workers because of high training costs
- When quasi-fixed costs are high, firms prefer to retain or hire trained workers
- Leads to a segmentation into protected and unprotected workers

Also leads firms to work employees more intensively rather than hire new workers

> **Example**
>
> Big law firms work associates long hours instead of hiring new associates

Work Sharing

Some individuals would embrace work sharing

- Two or more people share a job
- Hire more people to each work less hours for same total hours

⚠ Why Firms Resist

Quasi-fixed costs make it costly to bring on additional workers — firms prefer fewer workers working more hours, even if total hours worked is the same

Work Sharing

Introduction

Canada's Work Sharing Program

Government program that subsidizes employee wages during downturns, administered through Employment Insurance

How it works:

- Allows firms to reduce hours instead of laying workers off
- Employees form a "work sharing group" and agree to reduce hours
- Maximum duration of 26 weeks with possible extensions

The key idea is to avoid layoffs and employ more people

Introduction

Other possible work sharing schemes:

- Delayed entry into the labour force
- Early retirement
- Short work weeks
- Reduced overtime
- Increased vacation time

Key Question

Do any of these schemes actually increase the number of jobs?

Overtime Restrictions

One way to share work is to restrict overtime

- Some governments have enacted legislation to prevent excessive overtime
- Firms do not like this because it could increase costs

It is not clear that these laws increase employment:

- Firms may not hire more workers
- Increased costs reduce overall demand for labour
- New workers may not be as productive as existing workers
- There may not be enough employable workers available

Overtime Restrictions

⚠ Lump of Labour Fallacy

The belief that there is a fixed number of jobs in the economy — leading people to think employees working longer hours take jobs away from others

In reality:

- The number of jobs is not fixed
- As people become employed, the economy grows and more jobs are created
- Fallacy is commonly used to argue against immigration

ⓘ Evidence

Studies show that restricting work hours does not lead to increased employment — but may still be useful in the short term during economic downturns

Subsidized Work Sharing

Canada subsidizes work sharing programs for economic downturns

- Main goal: keep people employed during recessions
- Would be good if it also protects jobs longer term

Evidence

- Many firms use the program
- It does help keep people employed during downturns
- But little evidence that it increases employment longer term
    - Employees were laid off when subsidy ran out

Employee Demand for Work Sharing

Key Point

Employees need to also buy in for work sharing to be successful — it is not clear that they do in many cases

Example: union jobs

- Unions often resist work sharing programs
- Prefer to keep full-time jobs with benefits
- May see work sharing as a threat to job security
- May also see it as a way for employers to reduce wages and benefits

Summary

Summary

Non-wage benefits are common:

- Tax advantages, economies of scale, worker preferences, legal requirements
- Improve productivity and act as deferred compensation

Quasi-fixed costs complicate labour demand:

- Costs per worker independent of hours worked (hiring, training, termination, payroll taxes)
- Firm hires when present value of future productivity exceeds total costs
- Wage does not have to equal marginal product in each period

Summary

Quasi-fixed costs explain many labour market phenomena:

- Overtime, temp/gig workers, labour hoarding during downturns, market segmentation

Work sharing:

- Government programs help during recessions, but little evidence of longer-term employment gains
- Firms resist due to quasi-fixed costs; unions resist to protect full-time jobs